

## Parallelograms — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

### Objective

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- Q1 (c) **equal**. Opposite angles of a parallelogram are equal.
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- Q2 (b) **parallelogram**. Diagonals bisecting each other  $\Rightarrow$  parallelogram.
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- Q3 **115**. Adjacent angles are supplementary:  $180 - 65 = 115$ .
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- Q4 (b) **Diagonals equal**. Parallelogram diagonals bisect each other but are generally unequal.
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- Q5 **True**. A standard parallelogram property.
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- Q6 (a) **Both true, R explains A**. The converse property is exactly why the figure is a parallelogram.
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### Short answer

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- Q7  $\angle A = 72^\circ$ ,  $\angle B = 108^\circ$ .  $2x + 3x = 180 \Rightarrow x = 36$ .
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- Q8  $y = 105^\circ$ ,  $z = 75^\circ$ .  $y$  is adjacent to  $75^\circ$  (supplementary);  $z$  is opposite to  $75^\circ$  (equal).
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- Q9 **PR = 8 cm, QS = 6 cm**. Diagonals bisect each other, so  $PR = 2 \times PO$  and  $QS = 2 \times QO$ .
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### Long answer

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- Q10  $45^\circ, 135^\circ, 45^\circ, 135^\circ$ .  $x + 3x = 180 \Rightarrow x = 45$ ; opposite angles equal.
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- Q11  $\angle B = 100^\circ$ ,  $\angle C = 80^\circ$ ,  $\angle D = 100^\circ$ .  $\angle C = \angle A$  (opposite);  $\angle B = 180 - 80$  (adjacent);  $\angle D = \angle B$  (opposite).
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- Q12 **Fourth angle =  $110^\circ$ ; not a parallelogram**.  $360 - (90 + 75 + 85) = 110$ ; opposite angles ( $90^\circ$  &  $85^\circ$ ,  $75^\circ$  &  $110^\circ$ ) are not equal, so it cannot be a parallelogram.
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Total: 21 marks.